Module 06: Learning — Bayesian Model Selection and Structure Learning

> **Course**: Active Inference 401 | **Unit**: Advanced Theory | **Audience**: Graduate students / researchers

Learning Objectives

1. Analyze **Bayesian Model Selection** — choosing between competing generative models based on their evidence.
2. Evaluate **Bayesian Model Reduction** (BMR) as a computationally efficient method for pruning model parameters.
3. Examine **structure learning** — discovering the topology of the generative model (which variables and connections exist).

Key Concepts

1. Bayesian Model Selection (BMS)

Bayesian Model Selection compares models not on fit alone, but on **model evidence** — the marginal likelihood:

$p(o \mid m) = \int p(o \mid \theta, m)\, p(\theta \mid m)\, d\theta$

Key properties:

- Model evidence automatically balances fit against complexity (Occam's Razor)
- More complex models fit the data better but are penalized for their larger parameter spaces
- The "Occam window" is the range of model complexity where evidence is maximized

Bayes factors: The ratio of model evidences quantifies relative support:

$BF_{12} = p(o \mid m_1) / p(o \mid m_2)$

BF > 3: moderate evidence for m_1. BF > 20: strong evidence. BF > 150: very strong evidence.

Free energy approximation: Since model evidence is intractable, free energy F provides a lower bound:

$$\ln p(o \mid m) \geq -F[q, m]$$

Model selection using F is equivalent to selecting the model with lowest (most negative) free energy.

2. Bayesian Model Reduction (BMR)

BMR is a computationally elegant method for comparing reduced models without re-fitting them:

The idea: Given a full model m_0 with posterior $q(\theta \mid o, m_0)$, compute the evidence for a reduced model m_1 (with some parameters fixed or removed) *analytically* from the full model's posterior, without re-running inference.

The formula: Under the Laplace approximation:

$$\ln p(o \mid m_1) \approx \ln p(o \mid m_0) + \ln p(\theta \mid m_1) - \ln p(\theta \mid m_0) - \tfrac{1}{2} \mathrm{tr}[(\Sigma_1^{-1} - \Sigma_0^{-1}) \Sigma\_posterior]$$

where θ* is the MAP estimate and Σ are prior covariances. This is a one-step computation — no iterative optimization.

Applications:

- **Sleep-dependent pruning**: BMR formalizes how sleep could simplify the generative model by testing which parameters can be removed without significantly reducing evidence
- **Anatomical homology**: Compare brain region models with and without specific connections
- **Clinical diagnosis**: Compare disease models with and without specific pathological parameters

3. Structure Learning

Structure learning goes beyond parameter estimation to discover the model's **graph structure** — which variables exist and how they connect:

Three approaches:

1. Score-based: Enumerate possible structures, compute model evidence for each, select the best. Computationally expensive (exponential in number of variables).

2. Constraint-based: Use conditional independence tests to identify the structure. If $X \perp Y \mid Z$, then there's no direct edge between X and Y given Z. More computationally efficient but sensitive to test reliability.

3. BMR-based: Start with a fully connected model and use BMR to progressively remove edges (connections) that don't contribute to model evidence. This is the approach most consistent with Active Inference.

Dirichlet process priors: For discovering the number of hidden states (not just their connections), Bayesian nonparametric methods place priors over the model structure itself:

p(number of states) = Dirichlet process

This allows the model to "grow" new states when existing states are insufficient.

4. Hierarchical Bayesian Models

Hierarchical models treat parameters of one model as generated by a higher-level model:

Two-level hierarchy:

- Level 1: $p(o \mid \theta, m)$ — data given parameters
- Level 2: $p(\theta \mid \varphi, m)$ — parameters given hyperparameters
- Level 3: $p(\varphi \mid m)$ — hyperparameters given model

Empirical Bayes: Estimate hyperparameters φ by maximizing the marginal likelihood at the second level. This provides a principled method for setting priors — they are learned from the data.

Parametric Empirical Bayes (PEB): In group studies, PEB treats individual subjects' parameters as drawn from a group-level distribution. This implements "borrowing strength" — individuals with less data are regularized toward the group mean.

5. Model Space and Exploration

The space of possible models is vast. Efficient exploration requires:

Model space topology: Define a neighborhood structure on models — model m_1 is a neighbor of m_2 if they differ by one parameter or connection. Structure learning explores this space.

Greedy search: Start with a full model, iteratively remove the least evidenced parameter (BMR). Stop when no further reduction improves evidence. This is computationally tractable for moderate-sized models.

Reversible jump MCMC: For larger model spaces, RJMCMC samples across models of different dimensions, proposing additions and deletions of parameters. This explores the full posterior over model structures.

Summary

Bayesian Model Selection compares generative models on their evidence, automatically balancing fit against complexity. BMR provides a computationally efficient way to prune parameters without refitting. Structure learning discovers the graph topology of the generative model. Hierarchical models and empirical Bayes enable learning at the meta-level — learning how to learn.

Further Reading

- Friston, K. J. & Penny, W. D. (2011). Post hoc Bayesian model selection. *NeuroImage*, 56(4), 2089-2099.
- Friston, K. J. et al. (2018). Bayesian model reduction. *arXiv preprint* arXiv:1805.07092.
- Henson, R. N. et al. (2016). A Parametric Empirical Bayesian framework for fMRI-constrained MEG/EEG source reconstruction. *Human Brain Mapping*, 37(3), 1172-1186.